Hugging Robot

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Abstract - **Robots are increasingly becoming a part of our daily lives. From therapeutic robots to machinery to everyday technology the dependency and importance of robots in our everyday lives is becoming more and more visible. It is no wonder that in our lives and media we can see so many controversial discussions about robots and their place in the future. We can often see media regarding in the future robots gaining consciousness and revolting against humans and humans mistreating robots etc. Tommy the hugging robot is a robot that is not only a robot that "cures with love", but also it serves as an alternative purpose for people such as dementia patients to have a companion.**

I. Design

When originally thinking of the design of a hugging robot, the logical idea was to create a human-like robot. So when dementia patients or other elders miss a loved one or mistake someone as the loved one they can hug the robot and talk to the robot as if it were the loved one. When actually constructing the robot and researching other similar robots, then the fallacies of this idea began to appear.

A more realistic and human-like robot would be creepy, and frightening for the patient and anybody. The reasoning behind this is the uncanny valley effect [1]. When a nonhuman object shows human characteristics up to a certain point, then it is accepted, but past that point creates the opposite effect. Therefore creating a human-like hugging robot would create the opposite effect of what we are going for with the hugging robot. Having a realistic human-looking robot with unhumanlike [robotic] movements would be falling into this uncanny valley effect.

If the human-like look for the robot would be uncomfortable for patients, then the robotic look would be pleasing for them. According to Johnson, the robots would have their own form, one designed for whatever work they were meant to do, and the disconnect between our expectations and their appearance wouldn't occur. The best thing for our hugging robot would then be to take its own form therefore we made it look like a very stereotypical robot. Since it is a hugging robot, it should be large enough to hug a human.

To create a life-size robotic looking robot. People's average shoulder width was measured giving some extra space in case the people were wearing jackets. The torso of the robot is large enough to receive a hug from a human.

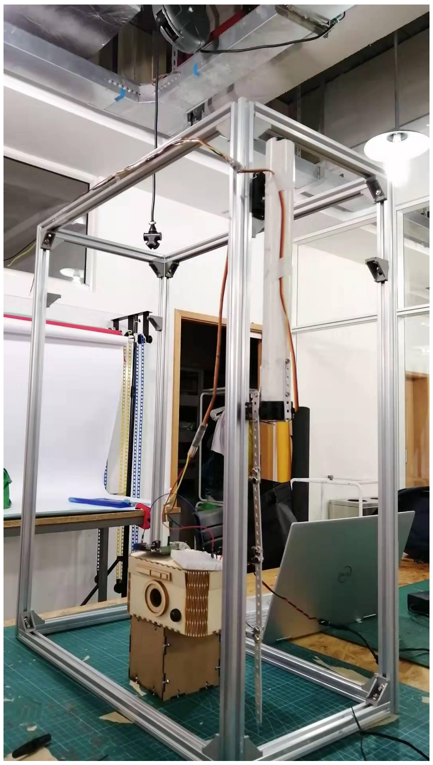

Figure 1: Metal Framing of Robot Torso

The metallic frame was selected not only for sturdiness, but also to create that robotic look and framing. A clear glass cover for the front was selected to be able to see the wiring and "robotic" insides which would add on to the robotic look.

II. Hugging Motion

The robot was programmed using an arduino to power the four motors. First the two shoulder motors were to move the physical arm up vertically. The second two motors placed at the "elbow" of the robot moved vertically to finish the hugging motion.

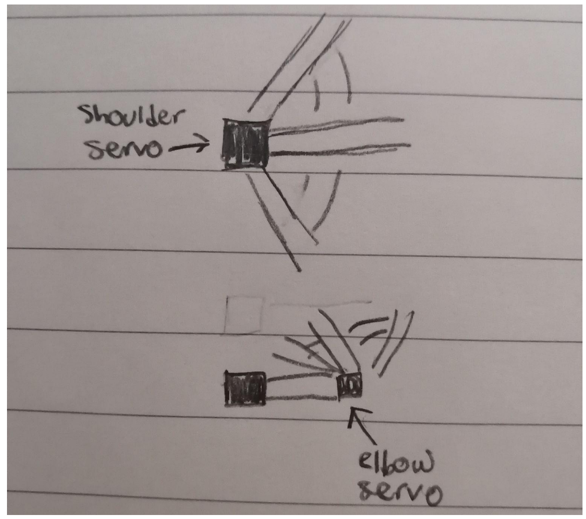

Figure 2: Diagram of servo motors

Creating the hugging motion theoretically isn't difficult, but working with the servos and creating a fluid hugging motion is quite difficult. Especially when there is the same code being processed, but the robot does not perform the same task every single time.

Originally the motors used for the robot were not strong and powerful enough which also caused trouble especially for the shoulders since the shoulders must carry the entire arm. After switching to a more powerful motor, it solved several problems.

References

1. Johnson, B. (2020, August 2). *What makes realistic robots so creepy?* HowStuffWorks Science. Retrieved December 17, 2021, from https://science.howstuffworks.com/realistic-robots-creepy.htm
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